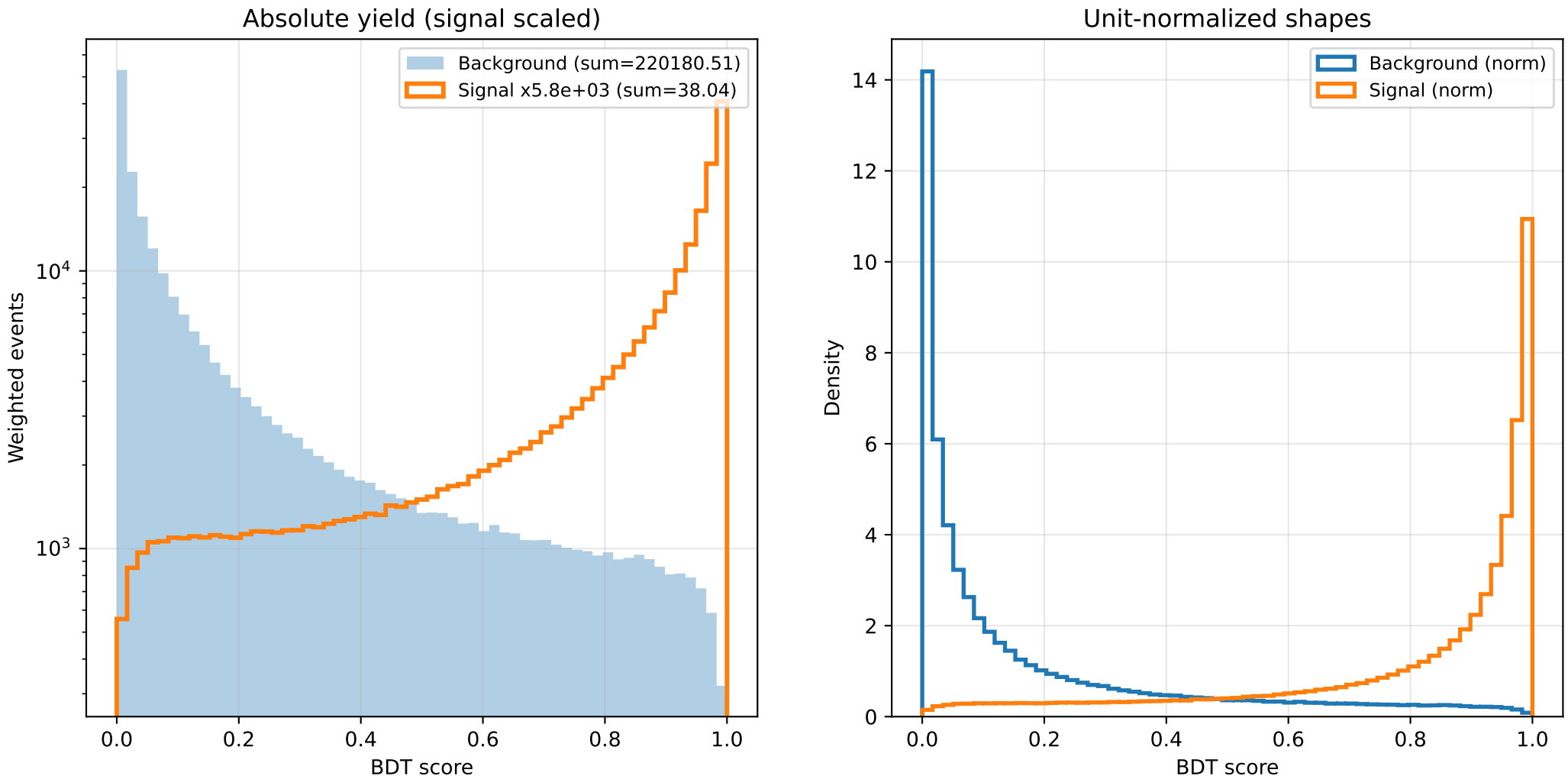
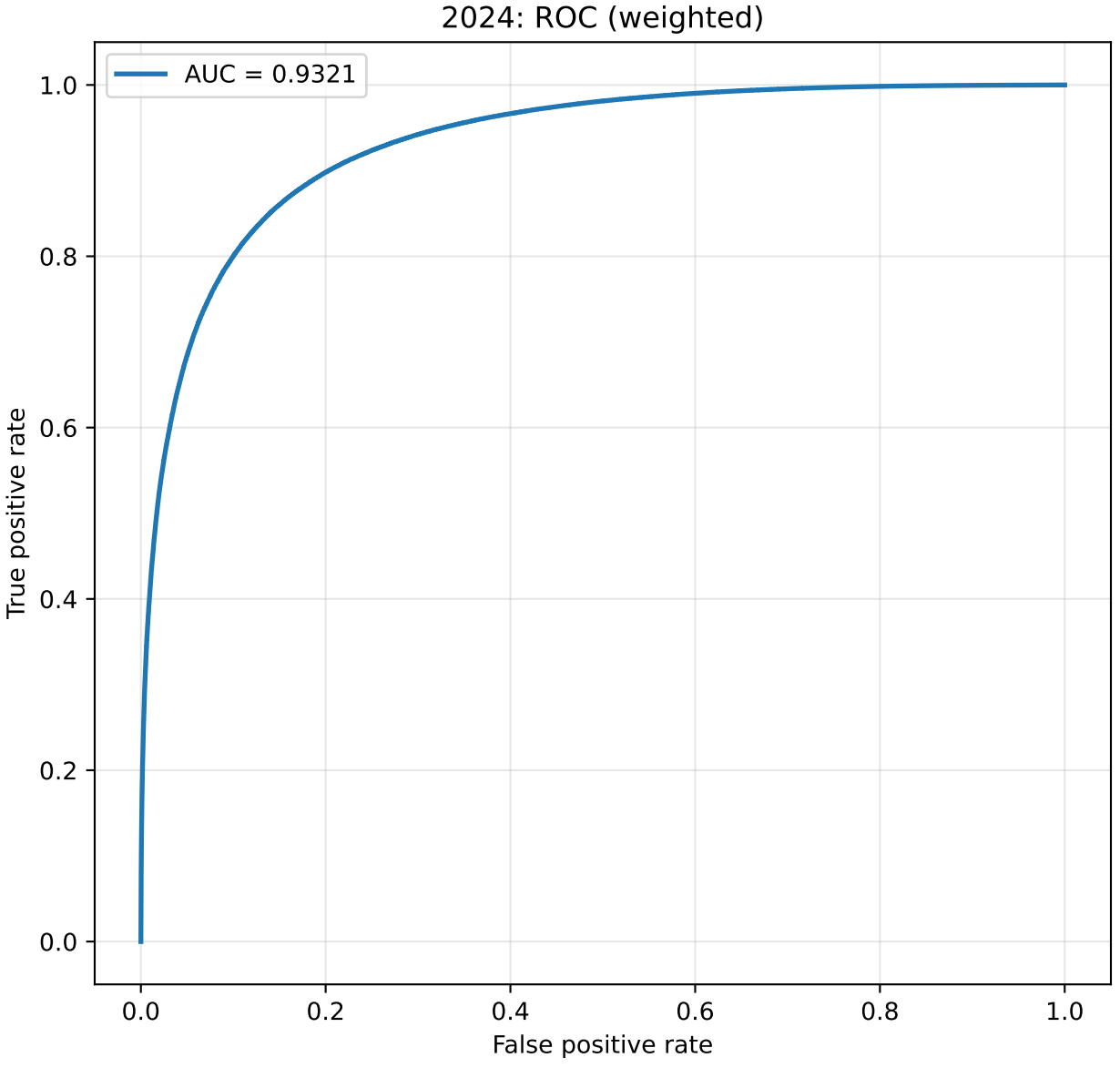
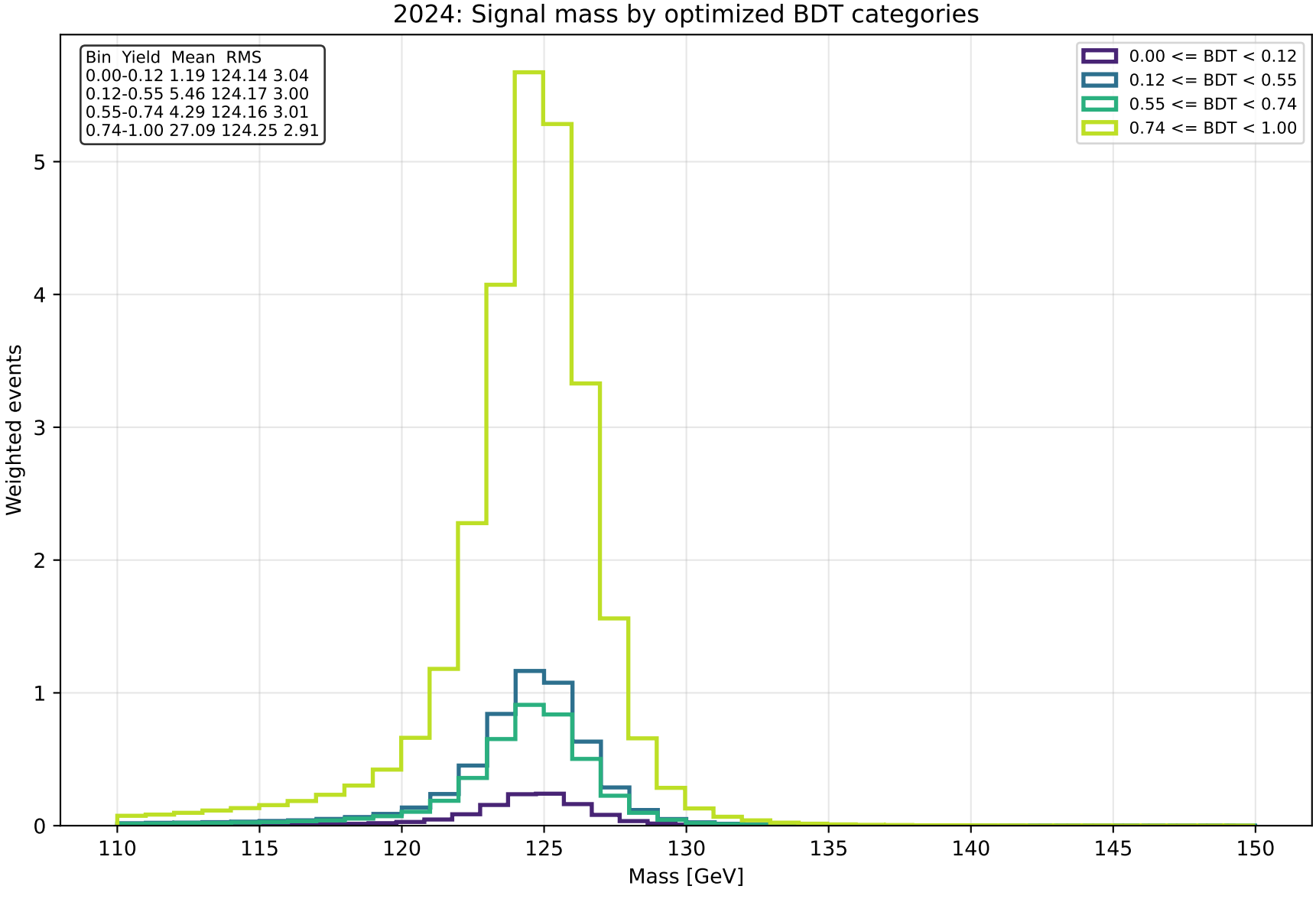
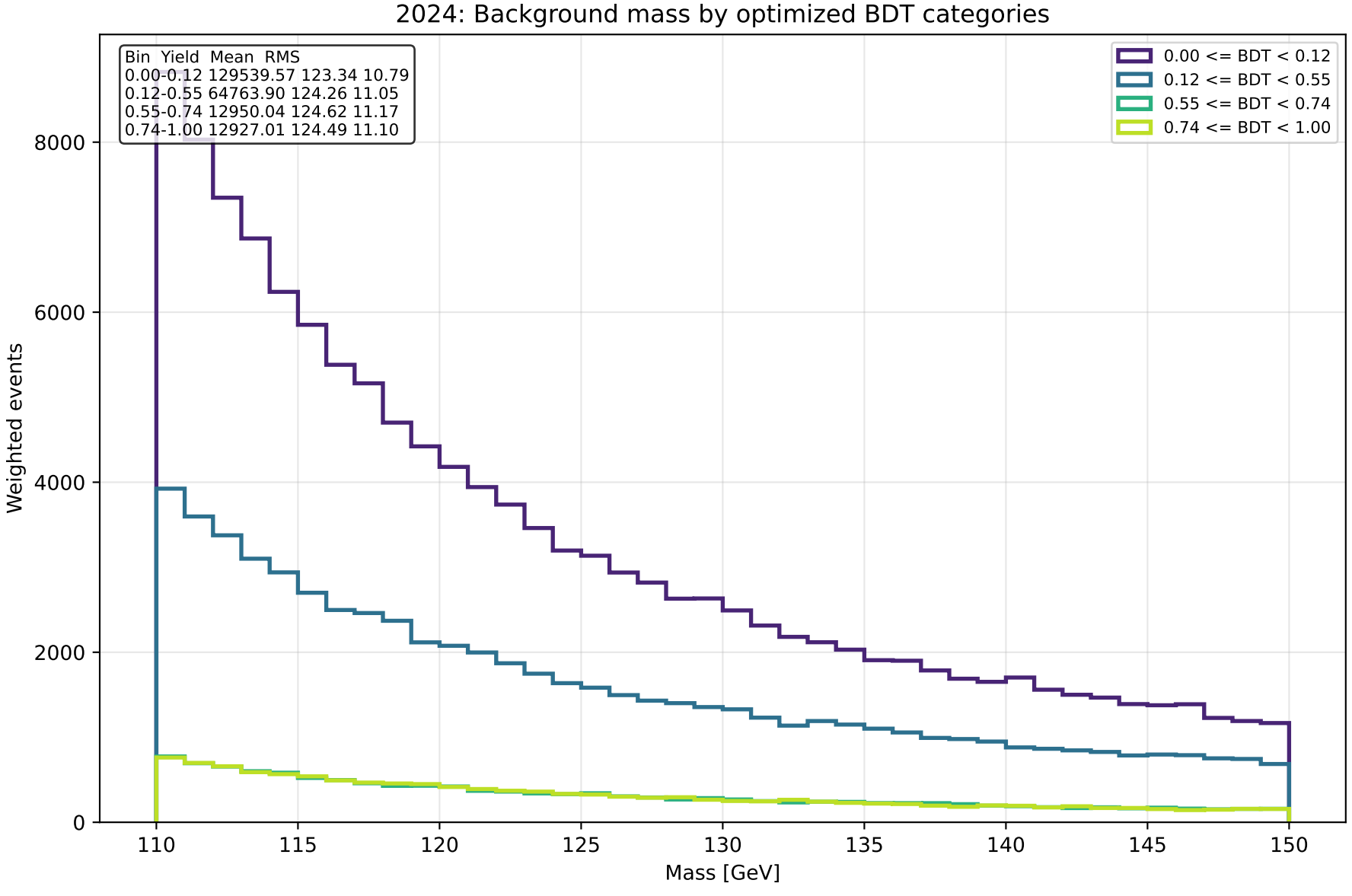

2024: BDT scores in 110-150 GeV
Absolute yield (signal scaled)
Background (sum=220180.51)
Signal x5.8e+03 (sum=38.04)
Weighted events
10^4
10^3
0.0
0.2
0.4
0.6
0.8
1.0
BDT score
Unit-normalized shapes
Background (norm)
Signal (norm)
Density
0
2
4
6
8
10
12
14
0.0
0.2
0.4
0.6
0.8
1.0
BDT score

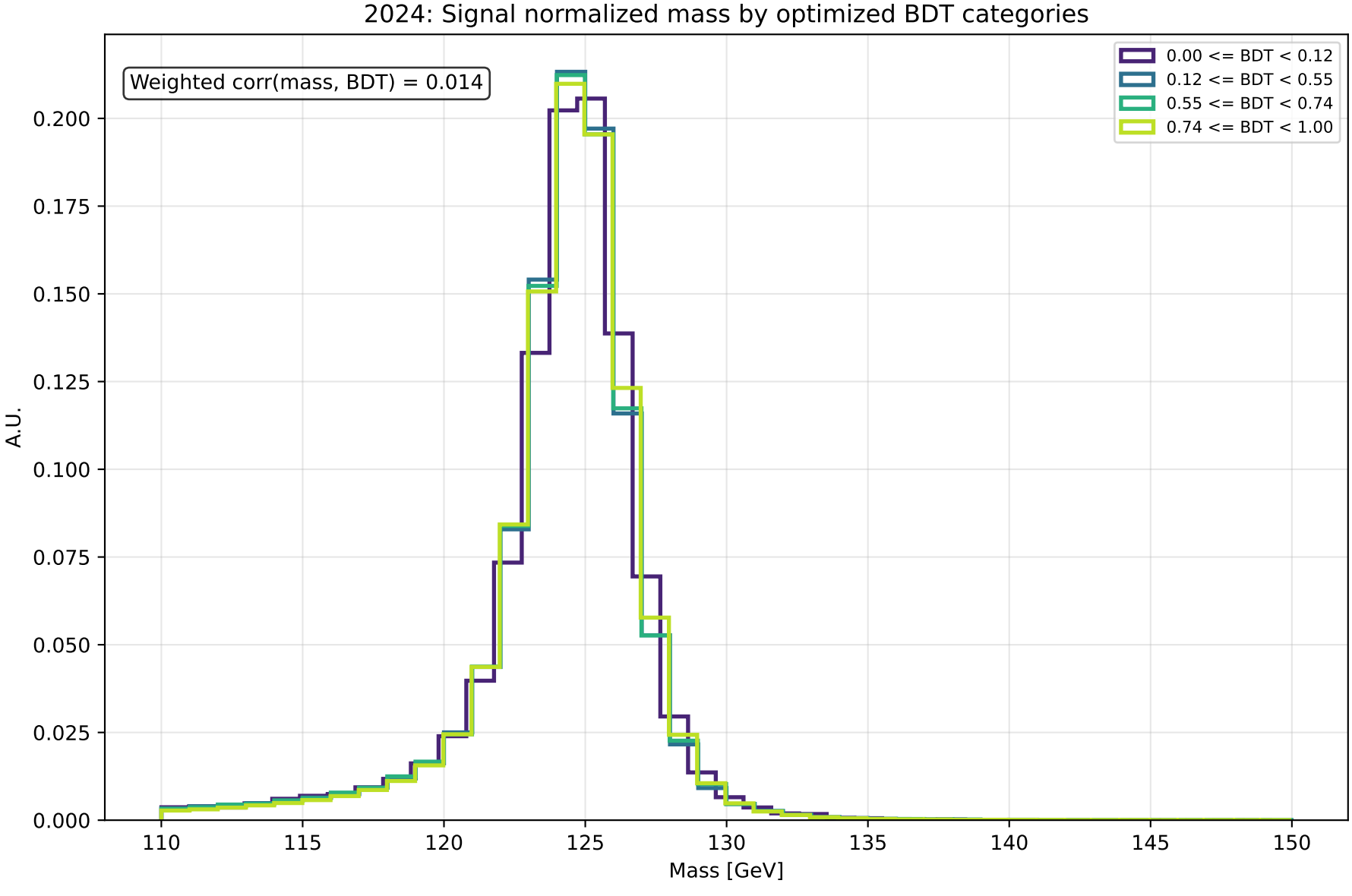

2024: ROC (weighted)
AUC = 0.9321
True positive rate
False positive rate
0.0
0.2
0.4
0.6
0.8
1.0

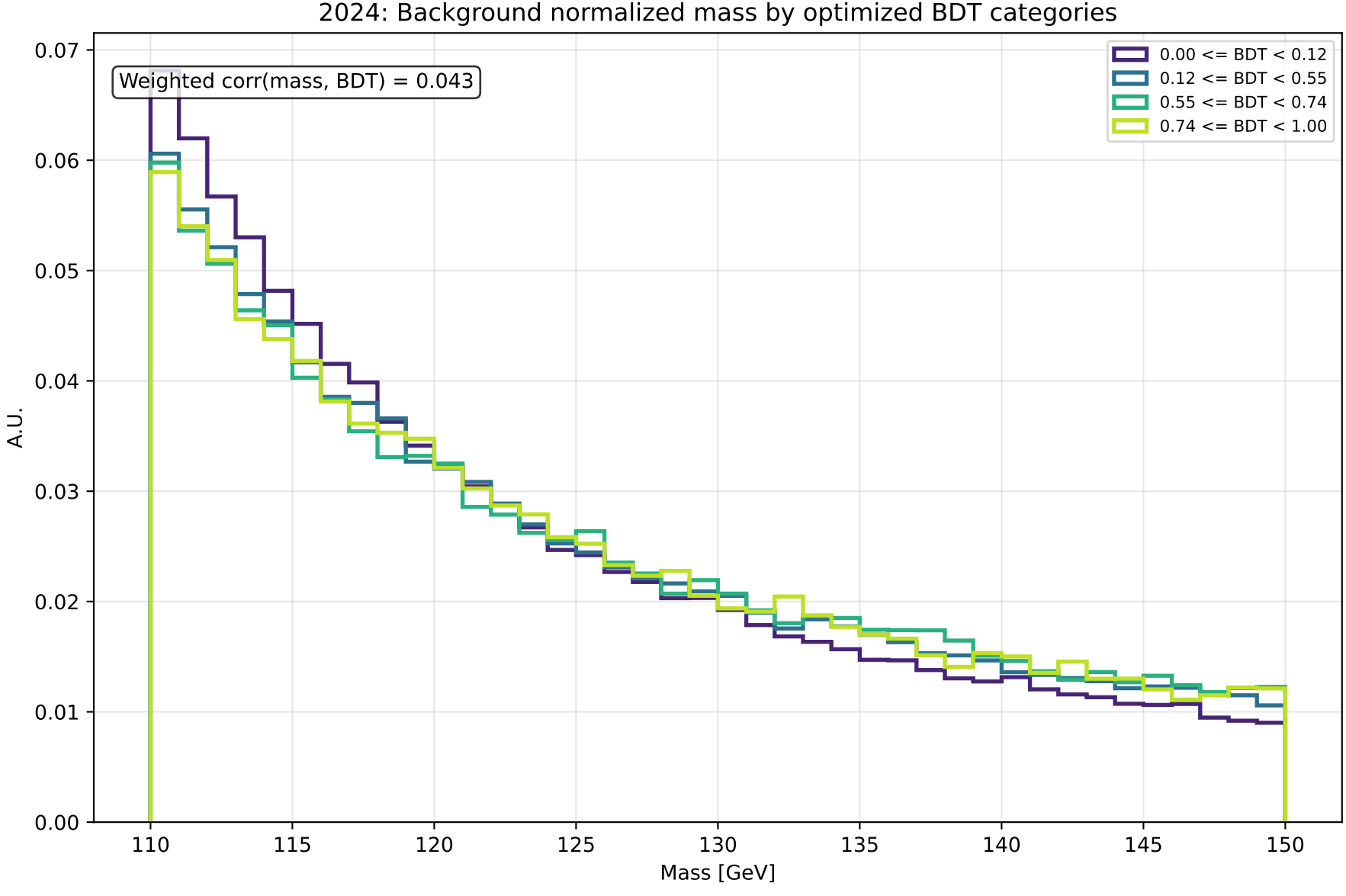
2024: Multi-bin BDT categorization (no hard cut)
Optimized N=4, Zexp(profile)=1.608, monotonic S/B=True

Bin	BDT range	S	B	S/B	S/sqrt(B)
1	0.000-0.122	1.194	129539.566	9.22e-06	0.003
2	0.122-0.546	5.465	64763.899	8.44e-05	0.021
3	0.546-0.739	4.288	12950.036	0.000331	0.038
4	0.739-1.000	27.094	12927.014	0.0021	0.238


2024: Signal mass by optimized BDT categories
Bin Yield Mean RMS
0.00-0.12 1.19 124.14 3.04
0.12-0.55 5.46 124.17 3.00
0.55-0.74 4.29 124.16 3.01
0.74-1.00 27.09 124.25 2.91
0.00 <= BDT < 0.12
0.12 <= BDT < 0.55
0.55 <= BDT < 0.74
0.74 <= BDT < 1.00
Weighted events
0
1
2
3
4
5
110
115
120
125
130
135
140
145
150
Mass [GeV]


2024: Background mass by optimized BDT categories
Bin Yield Mean RMS
0.00-0.12 129539.57 123.34 10.79
0.12-0.55 64763.90 124.26 11.05
0.55-0.74 12950.04 124.62 11.17
0.74-1.00 12927.01 124.49 11.10
0.00 <= BDT < 0.12
0.12 <= BDT < 0.55
0.55 <= BDT < 0.74
0.74 <= BDT < 1.00
Weighted events
0
2000
4000
6000
8000
110
115
120
125
130
135
140
145
150
Mass [GeV]


2024: Signal normalized mass by optimized BDT categories
Weighted corr(mass, BDT) = 0.014
0.00 <= BDT < 0.12
0.12 <= BDT < 0.55
0.55 <= BDT < 0.74
0.74 <= BDT < 1.00
A.U.
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.200
110
115
120
125
130
135
140
145
150
Mass [GeV]

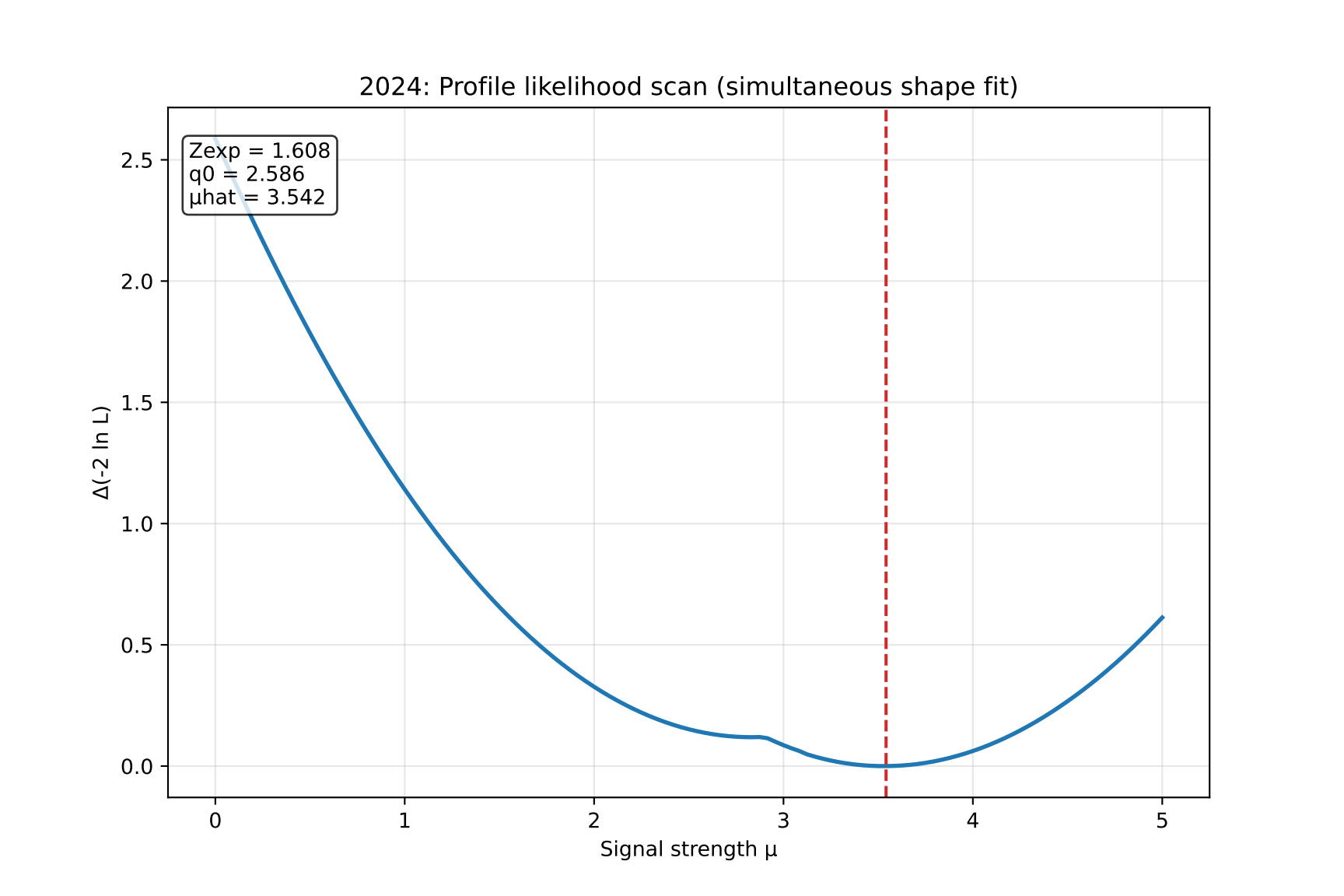

2024: Background normalized mass by optimized BDT categories
Weighted corr(mass, BDT) = 0.043
0.00 <= BDT < 0.12
0.12 <= BDT < 0.55
0.55 <= BDT < 0.74
0.74 <= BDT < 1.00
A.U.
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
110
115
120
125
130
135
140
145
150
Mass [GeV]

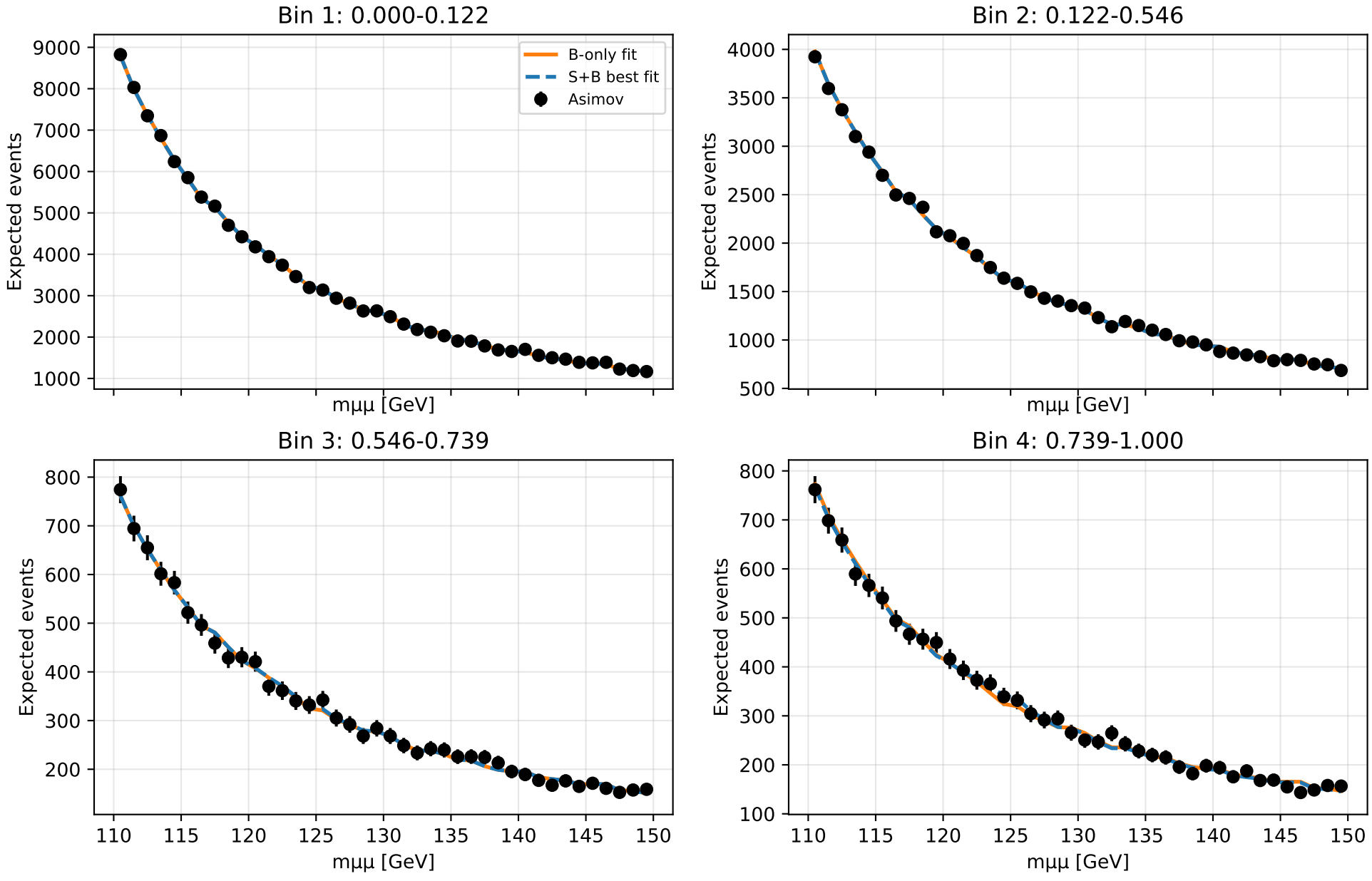
2024: Simultaneous shape-based profile likelihood summary

Quantity	Value
Expected Z	1.6082
q0	2.5864
mu_hat	3.5417
N_BDT bins	4
N_mass bins	40
poly order	1
bkg norm unc	0.200
poly unc	0.300
sig norm unc	0.050
sig mass shift unc [GeV]	0.200
sig resolution unc	0.100

2024: Shape-likelihood bin yields (Asimov)

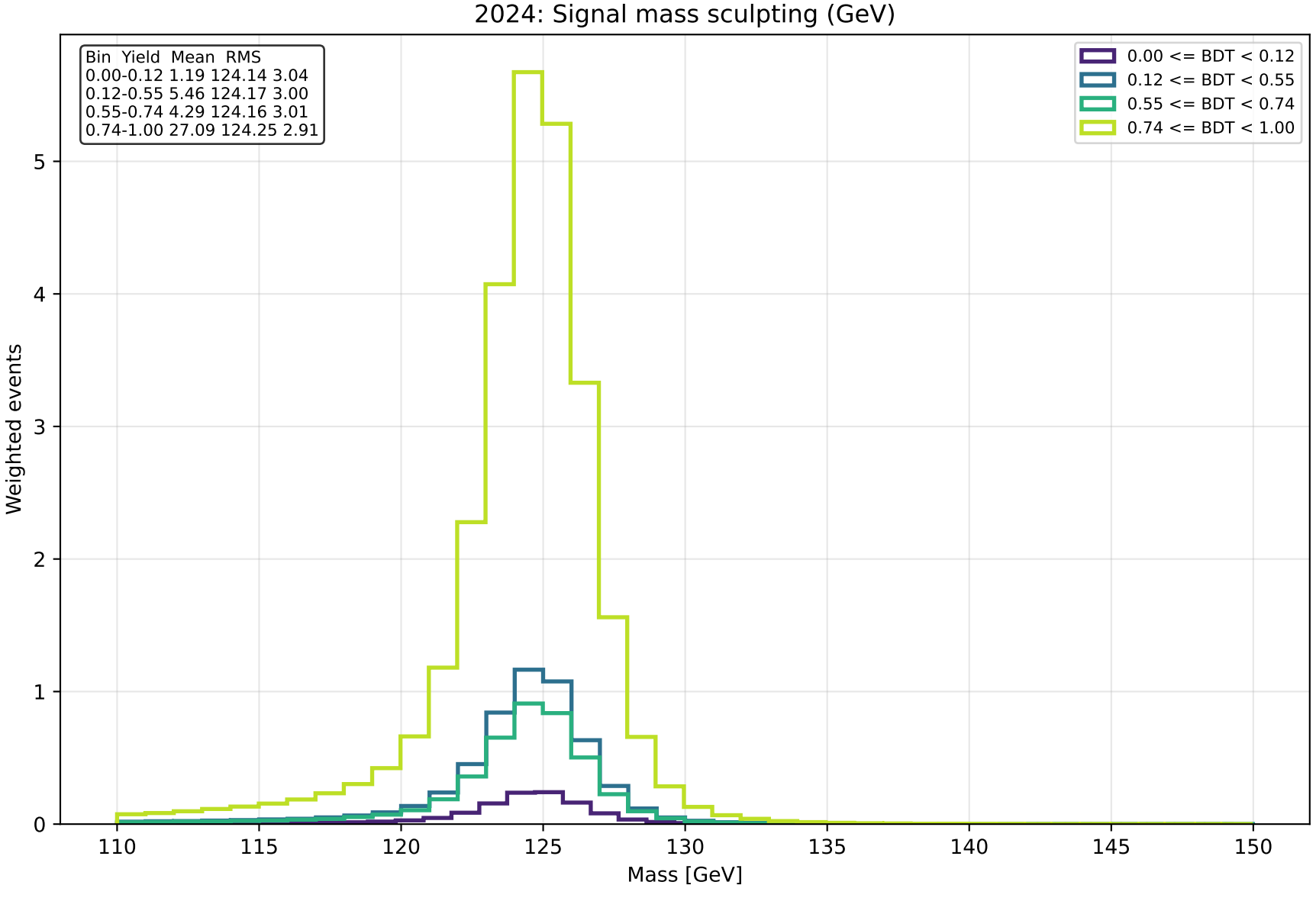
Bin	BDT	S truth	B truth	B-only fit	S+B fit
1	0.000-0.122	1.194	129539.566	129539.566	129543.796
2	0.122-0.546	5.465	64763.899	64763.899	64783.252
3	0.546-0.739	4.288	12950.036	12950.036	12965.223
4	0.739-1.000	27.094	12927.014	12927.014	12919.967

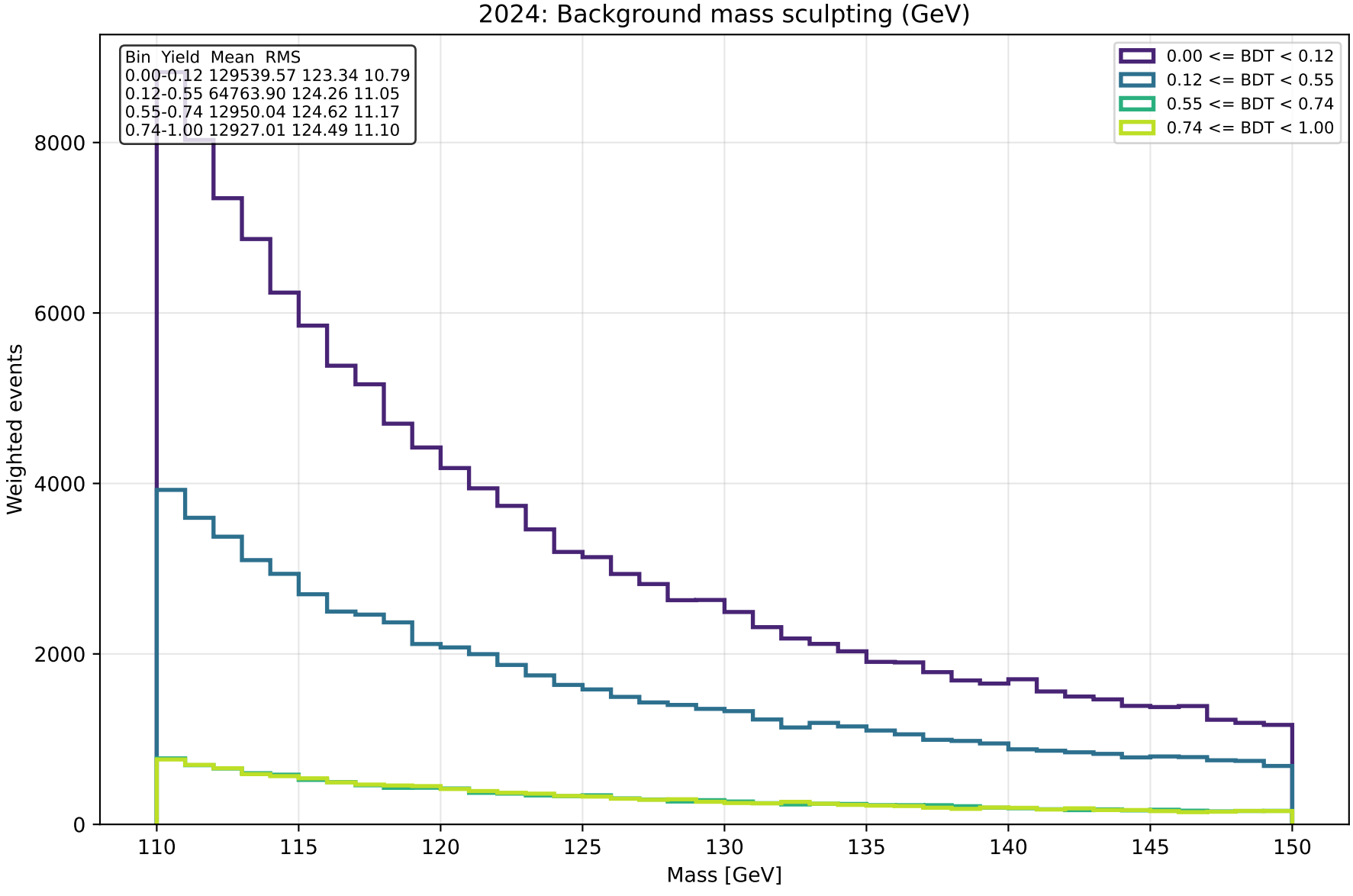

2024: Profile likelihood scan (simultaneous shape fit)
Zexp = 1.608
q0 = 2.586
μhat = 3.542
Δ(-2 ln L)
Signal strength μ
0.0
0.5
1.0
1.5
2.0
2.5
0
1
2
3
4
5

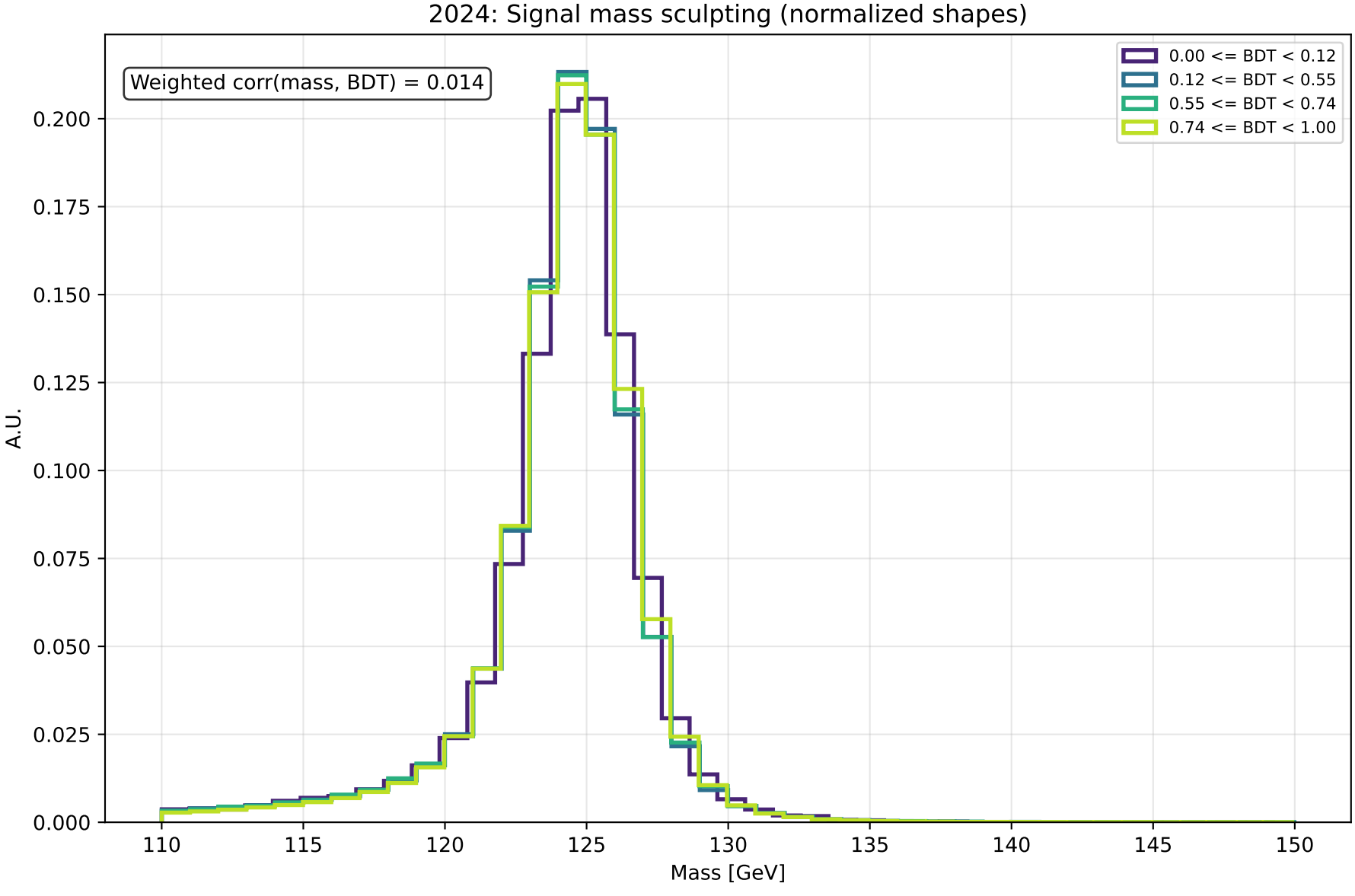

2024: Asimov mass templates by BDT bin (simultaneous fit)
Bin 1: 0.000-0.122
Bin 2: 0.122-0.546
Bin 3: 0.546-0.739
Bin 4: 0.739-1.000
B-only fit
S+B best fit
Asimov
Expected events
mμμ [GeV]

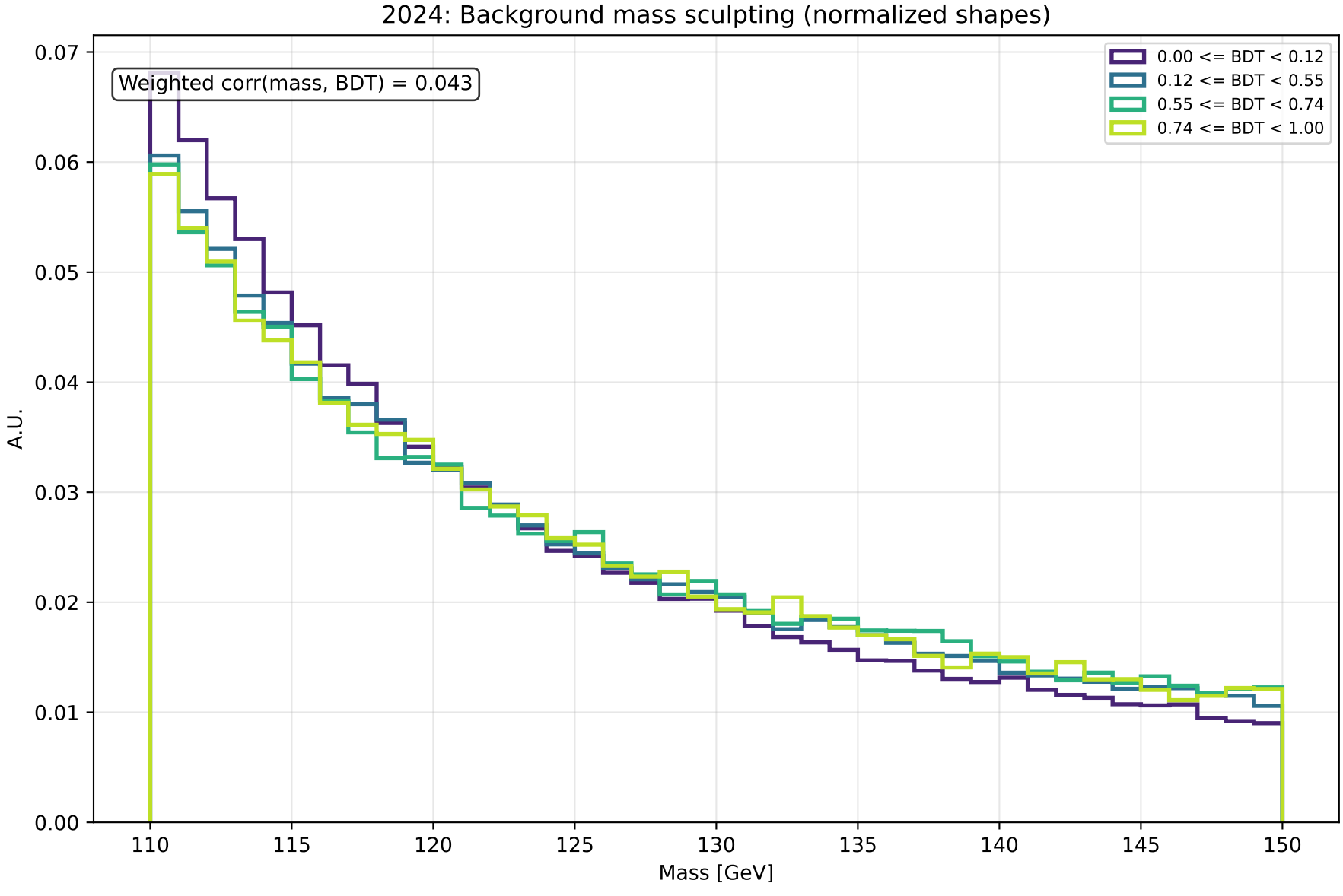
2024: Expected yield per optimized BDT bin (110-150 GeV)
S total=38.04, B total=220180.51

BDT bin	S (bin)	S frac	S>=lo	B (bin)	S/B (>=lo)
0.000-0.122	1.194	3.1%	100.0%	129539.566	0.000173
0.122-0.546	5.465	14.4%	96.9%	64763.899	0.000407
0.546-0.739	4.288	11.3%	82.5%	12950.036	0.00121
0.739-1.000	27.094	71.2%	71.2%	12927.014	0.0021

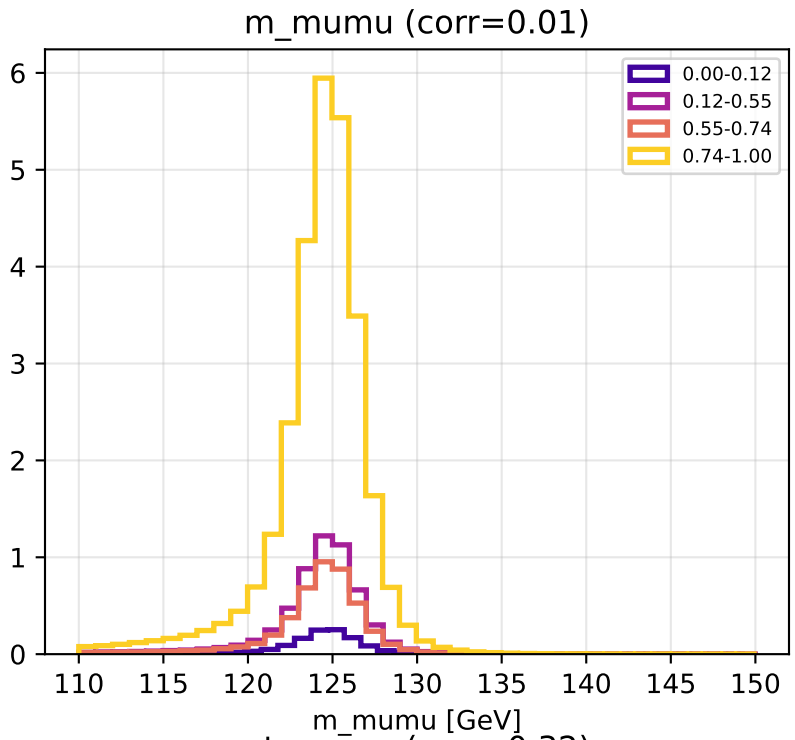

2024: Signal mass sculpting (GeV)
Bin Yield Mean RMS
0.00-0.12 1.19 124.14 3.04
0.12-0.55 5.46 124.17 3.00
0.55-0.74 4.29 124.16 3.01
0.74-1.00 27.09 124.25 2.91
0.00 <= BDT < 0.12
0.12 <= BDT < 0.55
0.55 <= BDT < 0.74
0.74 <= BDT < 1.00
Weighted events
0
1
2
3
4
5
110
115
120
125
130
135
140
145
150
Mass [GeV]

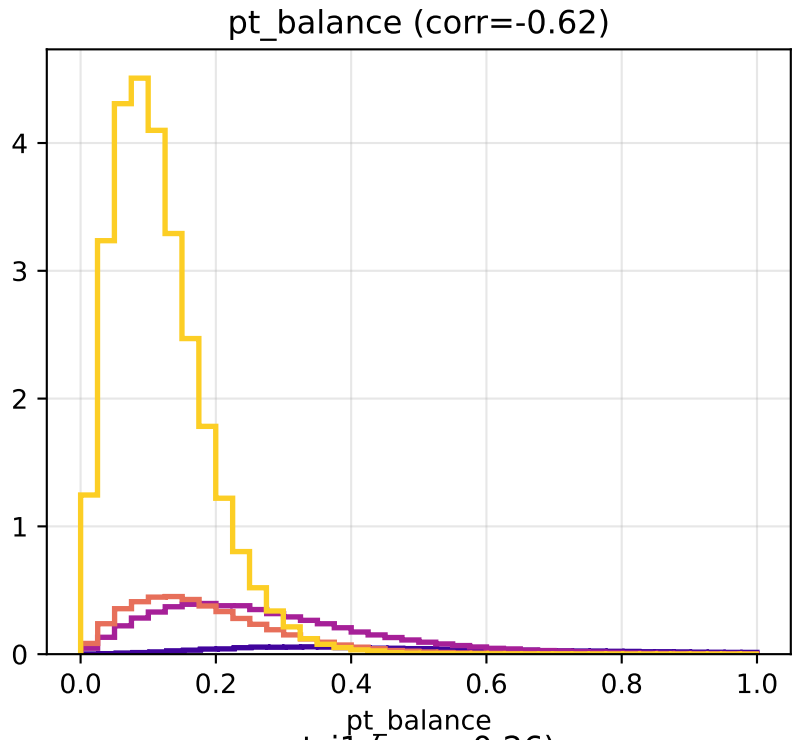

2024: Background mass sculpting (GeV)
Bin Yield Mean RMS
0.00-0.12 129539.57 123.34 10.79
0.12-0.55 64763.90 124.26 11.05
0.55-0.74 12950.04 124.62 11.17
0.74-1.00 12927.01 124.49 11.10
0.00 <= BDT < 0.12
0.12 <= BDT < 0.55
0.55 <= BDT < 0.74
0.74 <= BDT < 1.00
Weighted events
0
2000
4000
6000
8000
110
115
120
125
130
135
140
145
150
Mass [GeV]

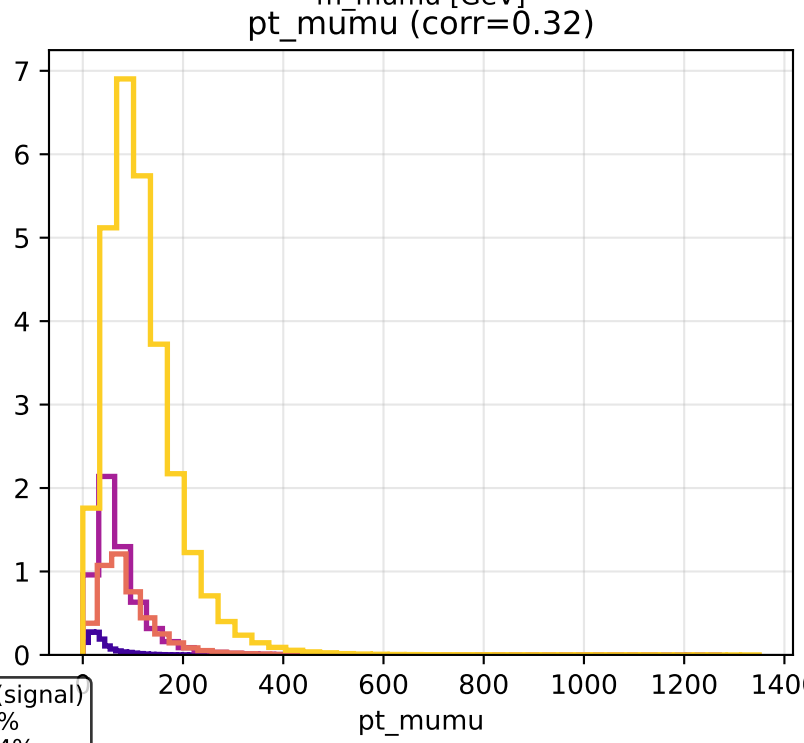

2024: Signal mass sculpting (normalized shapes)
Weighted corr(mass, BDT) = 0.014
0.00 <= BDT < 0.12
0.12 <= BDT < 0.55
0.55 <= BDT < 0.74
0.74 <= BDT < 1.00
A.U.
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.200
110
115
120
125
130
135
140
145
150
Mass [GeV]

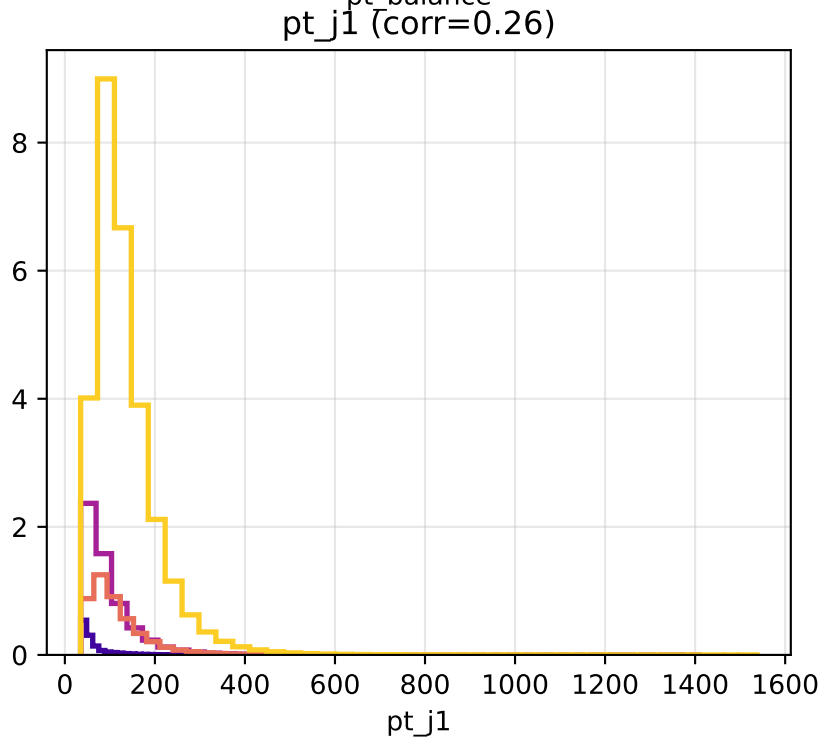

2024: Background mass sculpting (normalized shapes)
Weighted corr(mass, BDT) = 0.043
0.00 <= BDT < 0.12
0.12 <= BDT < 0.55
0.55 <= BDT < 0.74
0.74 <= BDT < 1.00
A.U.
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
110
115
120
125
130
135
140
145
150
Mass [GeV]

2024: Signal distributions by BDT bin

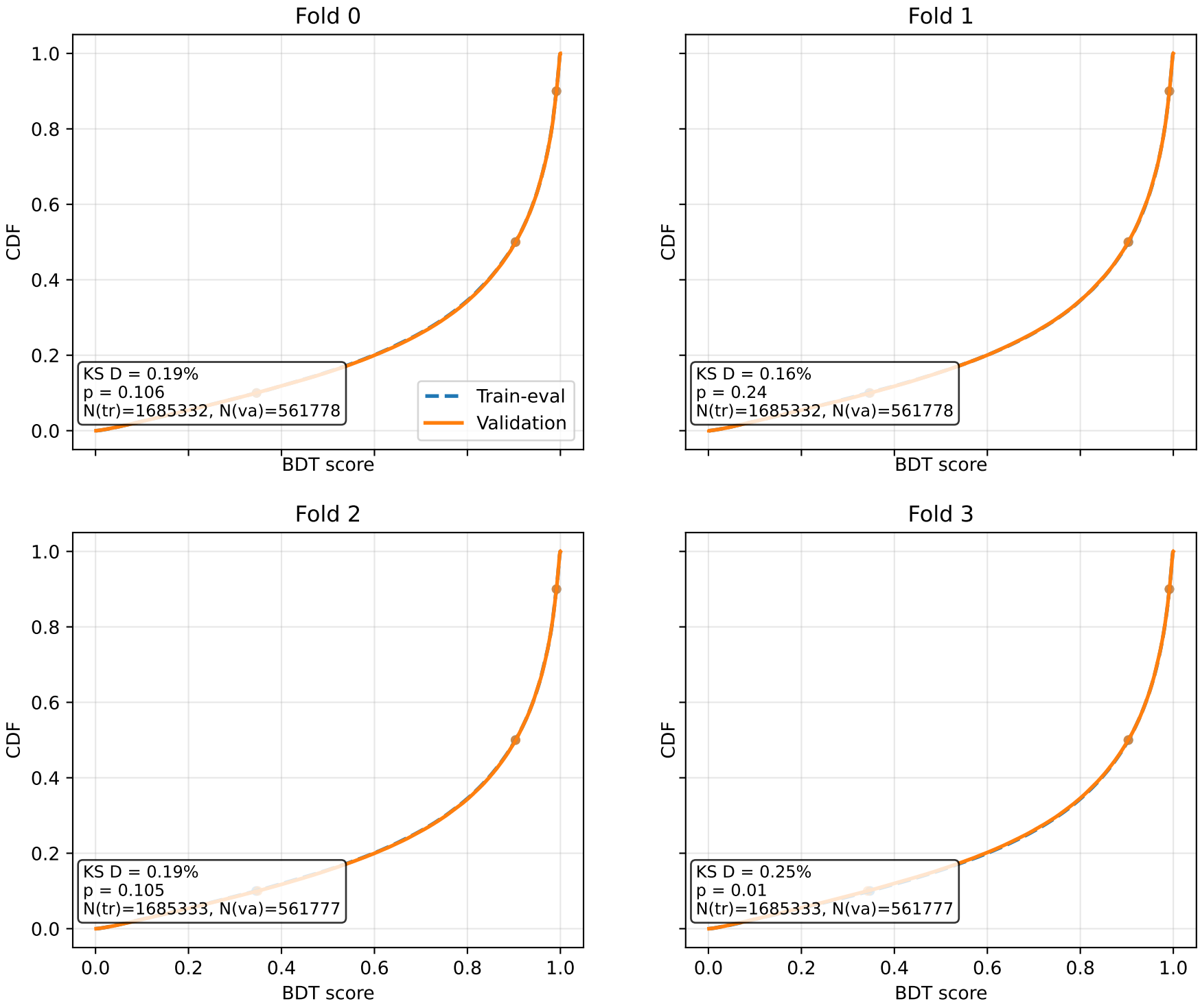








BDT bin yield (signal)
0.00-0.12: 3.1%
0.12-0.55: 14.4%
0.55-0.74: 11.3%
0.74-1.00: 71.2%


2024: CDF (train-eval vs validation)
Fold 0
KS D = 0.19%
p = 0.106
N(tr)=1685332, N(va)=561778
Train-eval
Validation
Fold 1
KS D = 0.16%
p = 0.24
N(tr)=1685332, N(va)=561778
Fold 2
KS D = 0.19%
p = 0.105
N(tr)=1685333, N(va)=561777
Fold 3
KS D = 0.25%
p = 0.01
N(tr)=1685333, N(va)=561777
CDF
BDT score
0.0
0.2
0.4
0.6
0.8
1.0

2024: KS test summary

Fold	KS statistic	N events	alpha	Pass threshold	KS pass
0	0.19%	561778	0.001	0.20%	TRUE
1	0.16%	561778	0.001	0.20%	TRUE
2	0.19%	561777	0.001	0.20%	TRUE
3	0.25%	561777	0.001	0.20%	FALSE

2024: Top-10 feature importance (gain)
2024_featureimportance_background.txt

Feature	Gain %
pt_balance	33.48%
pt_mumu	11.43%
pt_j1	11.10%
z	9.63%
dphi_mumu_jj	7.52%
deta_jj	7.02%
pt_j2	6.56%
njets	3.13%
jet1_eta	2.27%
mu1_pt_over_m	1.77%